

PROTOCOL FOR ASSESSING LIGHTING (INTERIOR/EXTERIOR)

Advanced Attribute Performance and Benchmarking
December, 2025





Problem Statement

Complaints

Vehicles meet Homologation requirements, yet customers report '*Poor Visibility*'. We need to measure this specific Delta.

Lack of Robust Vehicle Evaluation

Current lighting performance relies heavily on component-level specifications and carry-over concepts from previous projects.

Measurement Gap

We lack a standardized, repeatable procedure to convert "poor highway visibility" or "screen too bright" into objective integers.

Verification Deficit

Without an objective baseline, we cannot mathematically compare internal performance against benchmarks or verify if a fix actually works.





Project Objectives & Goals

Project Objectives

Standardize Evaluation

Develop a rigorous, vehicle-level testing procedure covering all use cases: highway range, foggy conditions, and HMI intrusiveness.

Measurement and Benchmarking

Use professional-grade sensors (Gossen, Bosch) to map the performance of competitor vehicles and set data-driven VTS targets.

Consultancy Alignment

Collaborate with M/s CATARC to ensure our internal metrics align with international laboratory standards and future homologation trends.

High-Level Goals

Robustness

Validate the procedure across the current fleet to ensure global engineering consistency.

Quality Defense

Move from "defending the design" to "proving the performance" through the cascading of Component -> System -> Vehicle.

Standardization

- **Subjective Vehicle Level:** "Good road visibility during nighttime."
- **Objective Vehicle Level:** "The beam hits 5 Lux at 120m."



VF Customer Complaints





Customer Feedback



VF3 – Headlight

- *KOL [Philippines, Indonesia]* Feedback: Reported insufficient intensity during unlit rural driving conditions.
- Technical Hypothesis: Possible deficiency in peak luminous flux or non-optimal beam aiming from the factory.



<https://vinfast.vn/dien-dan/thao-luan/loi-den-vf6-plus/>

VF6 - Headlight

Market Feedback: Erratic behavior of the dynamic low-beam leveling system (Headlight Leveling).

• **Symptom A (Visibility Loss):** While driving, the low beam automatically "dips" to an abnormally low position, significantly reducing forward visibility.

• **Symptom B (Braking Correlation):** The beam returns to the correct height only when the vehicle is braked or comes to a complete stop.

• **Symptom C (Glare Hazard):** Sometimes the beam raises too high and fails to auto-level, causing severe glare for oncoming traffic.



VF5 - Headlight

- *KOL Feedback:* Inconsistent brightness levels + frequently burn-out taillights.
- Technical Breakdown: VF5 might not provide the required peak candela (Lux at distance) to identify road hazards early enough for safe braking.



VF6 + VF7 HMI

- *KOL Observation:* HMI screen is "too bright" during night driving.
- Luminance Issue: Even when the brightness slider is set to the Lowest minimum, the light emitted is still high enough to cause significant eye strain during long night trips.
 - This suggests the display luminance is calibrated significantly higher than the comfort standard (**ISO 15008**).



ISO and ECE standard study





Identification of Core Metrics



Exterior standards:

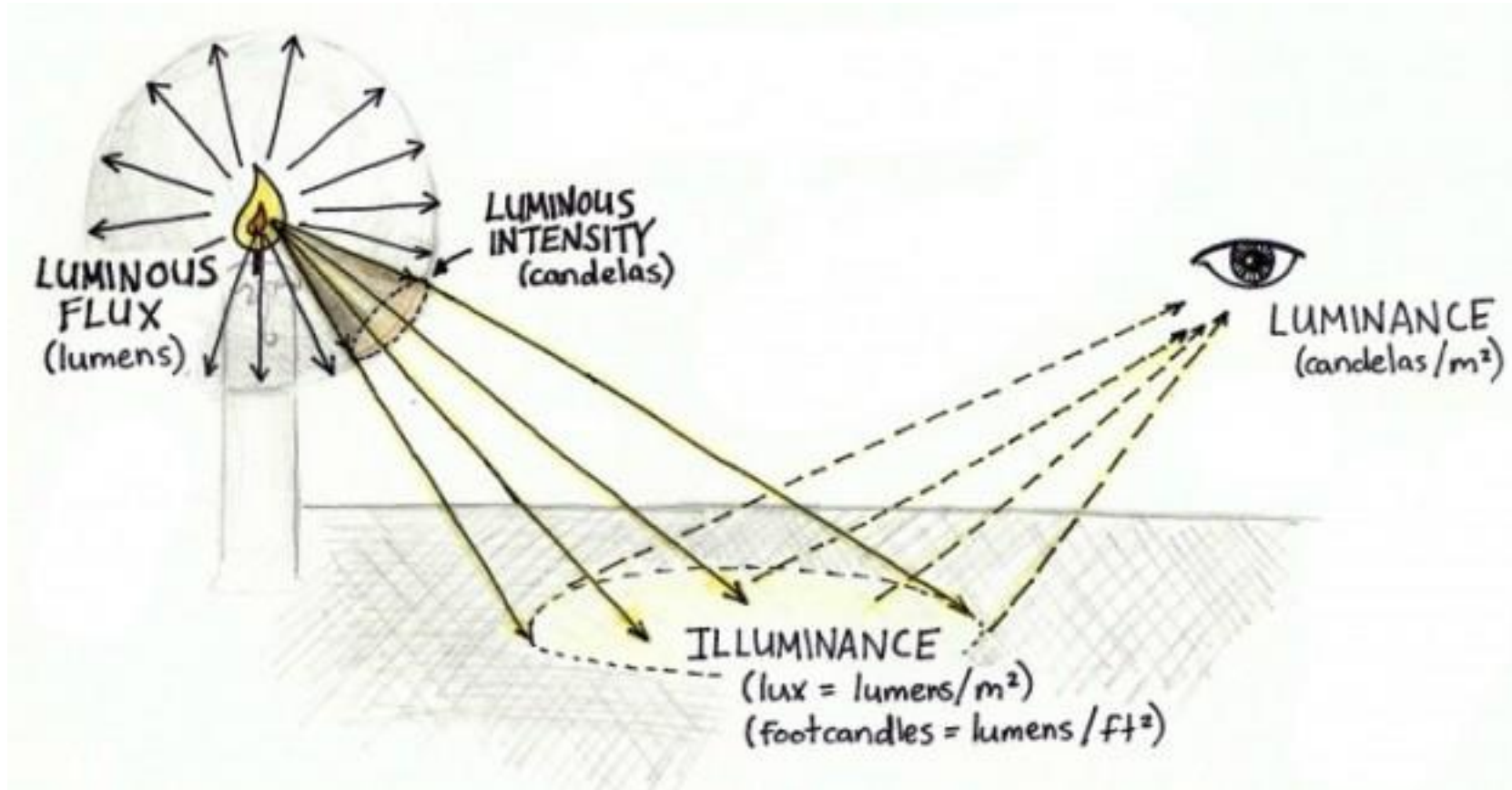
- UNECE R149: Headlamp performance.
- UNECE R150: Retro-reflectors.
- UNECE R148: Direction indicators.
- UNECE R48: Installation of lighting and light-signaling devices.

Interior Standards:

- ISO 15008: for in-vehicle visual displays
- ISO 15005: how lighting affects driver cognitive load

Disclaimer: Compliance with International Standards (UNECE/ISO) represents the legal minimum for market entry and **does not guarantee** satisfaction of customer performance expectations



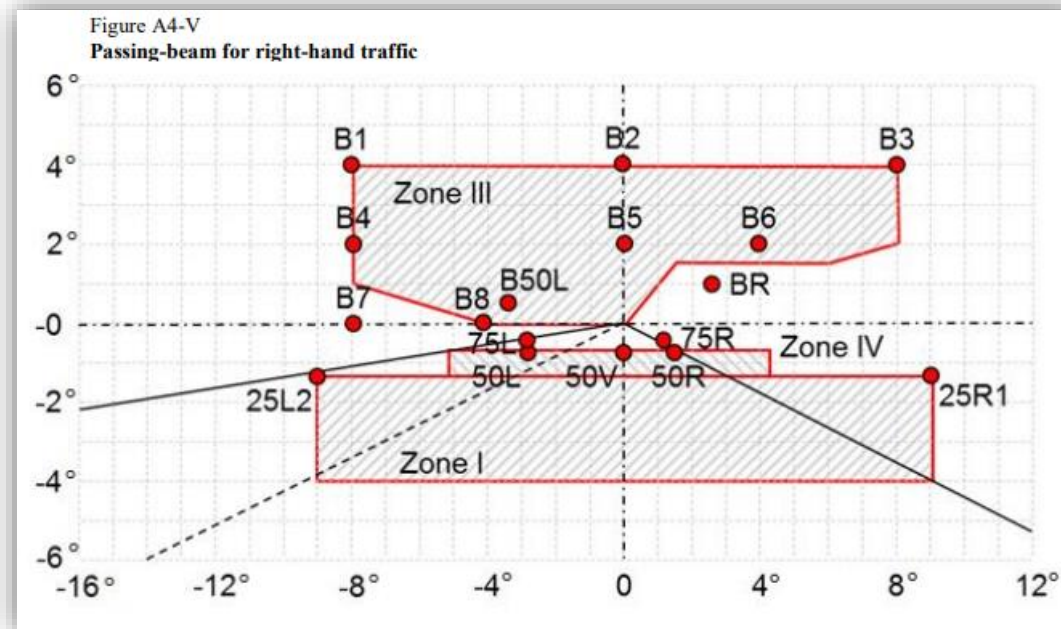




UNECE R149: Headlamp performance.

Analogy for Understanding Zones: If the light from your car is a **theatre stage**:

- **Zone III** is the **audience**: You want enough light to see them slightly, but never enough to blind them.
- **Zone IV** is the **center stage**: This is the most important part of the road where you need a strong, steady light to see your path clearly.
- **Zone I** is the **orchestra pit and floor**: It provides the general "wash" of light that lets you see the boundaries of the stage and what is right at your feet.





ISO 15008 (2017)

- Display illumination **should have a brightness control** to allow adjustment over a suitable range because of the very wide range of ambient illuminations that determine the driver's adaptation level.
- It defines the "**night condition**" as an ambient illumination of **less than 50 lx**.
- Night conditions: The document mandates a **minimum contrast ratio of 5:1** between the symbol and its background.
- During contrast measurements for the night condition, the display luminance is required to be adjusted to its "**night maximum**" setting.
- The standard recommends that **negative display mode** (Dark mode) should be used under the night condition.





Identification of Core Metrics

We transitioned from subjective descriptions (e.g., "poor visibility") to **Objective Integers**

Example:

Subjective Feedback (The Complaint)	Objective Metric	"Fail" Threshold	Source
"Headlights feel weak on highway"	Max Luminous Intensity (I_{max})	< 40,500 cd (Class B – for LED lights) < 10,100 cd (at point $75R = 1.15 \cdot R; 0.57 \cdot D$)	UN R149
"Dark spots in the beam pattern"	Beam Homogeneity (2D Lux Map)	Substantial non-uniformity (Visual gaps in Zones I, III, or IV)	UN R149
"Screen is too bright at night"	Minimum Luminance (Black Floor)	> 0.5 cd/m² (nits)	ISO 15008
"Glare from oncoming traffic"	Vertical Inclination (Cut-off)	> -0.5% (i.e., aimed larger than -0.5% down)	UN R48





Cross-Vehicle Benchmarking

We established a baseline by measuring benchmarks (**BMW / BYD**) under identical, controlled conditions.

To ensure data integrity, we utilize professional-grade sensors (**Bosch, Insize**) that remove the "human variable" from the evaluation.

Table: Benchmarking Isolation Protocols – Preparation Procedure Before Testing

Standardized procedures to ensure that VinFast vs. Competitor comparisons are mathematically valid.

Protocol Category	Engineering Specification	Standard / Basis
Voltage Control	Vehicles tested in "Ready Mode" at 13.5V ($\pm 0.1V$) to isolate battery drain.	UN R48
Optical Clarity	Lenses and sensors cleaned with standardized isopropyl wipe prior to test.	BMW Lab Protocol
Alignment Lock	Laser rangefinder and digital inclinometer used for 0° centerline calibration.	UN R149
Pitch Monitoring	Measurements invalidated if chassis pitch exceeds $\pm 0.3^\circ$ variance from average.	UN R149
Data Filtering	Photometer signals passed through a 35 Hz low-pass filter .	UN R149



VF5 Headlight – belongs to Class B

Table 1: Primary Photometric Control Points (Passing Beam - Class B)

These points are used to evaluate road visibility and glare prevention for asymmetrical beams.

Point	Angular Coordinates (H, V)	Req. (Min/Max cd)	Significance & Safety Role
75R	1.15° R, 0.57° D	10,100 (Min)	Right roadside visibility at ~75m; critical for detecting pedestrians.
50R	1.72° R, 0.86° D	10,100 (Min)	Right-side road illumination at closer range.
50V	0° (V-V), 0.86° D	5,100 (Min)	Center of the driving path; ensures core lane visibility.
B50L	3.43° L, 0.57° U	350 (Max)	Glare Protection: Simulates the eyes of an oncoming driver.
Zone III	Defined Polygon	625 (Max)	Prevents excessive light above the cut-off to avoid dazzling traffic.





VF6 & VF7 – HMI and Visual Presentation Targets (ISO 15008 / 15005)

Table: HMI and Visual Presentation Targets (ISO 15008 / 15005)

These metrics ensure display legibility and driver comfort across varying ambient light states.

Condition	Ambient Illuminance	Objective Metric	Target / Threshold
Direct Sunlight	10,000–50,000 lx	Contrast Ratio	> 3:1 (> 5:1 for safety-critical info).
Twilight	10–100 lx	Adaptive Dimming	Continuous adjustment via power function to prevent washout.
Night	< 50 lx	Luminance Floor	< 0.05 cd/m² (nits) to preserve dark adaptation.
Cognitive Load	Task-based	Glance Duration	Total glance time < 2.0 seconds per task.
HMI Usability	N/A	Dialogue Design	Must satisfy Controllability and Learnability principles.



Testing Methods Used By Other Manufacturers





BMW Light Channel Next

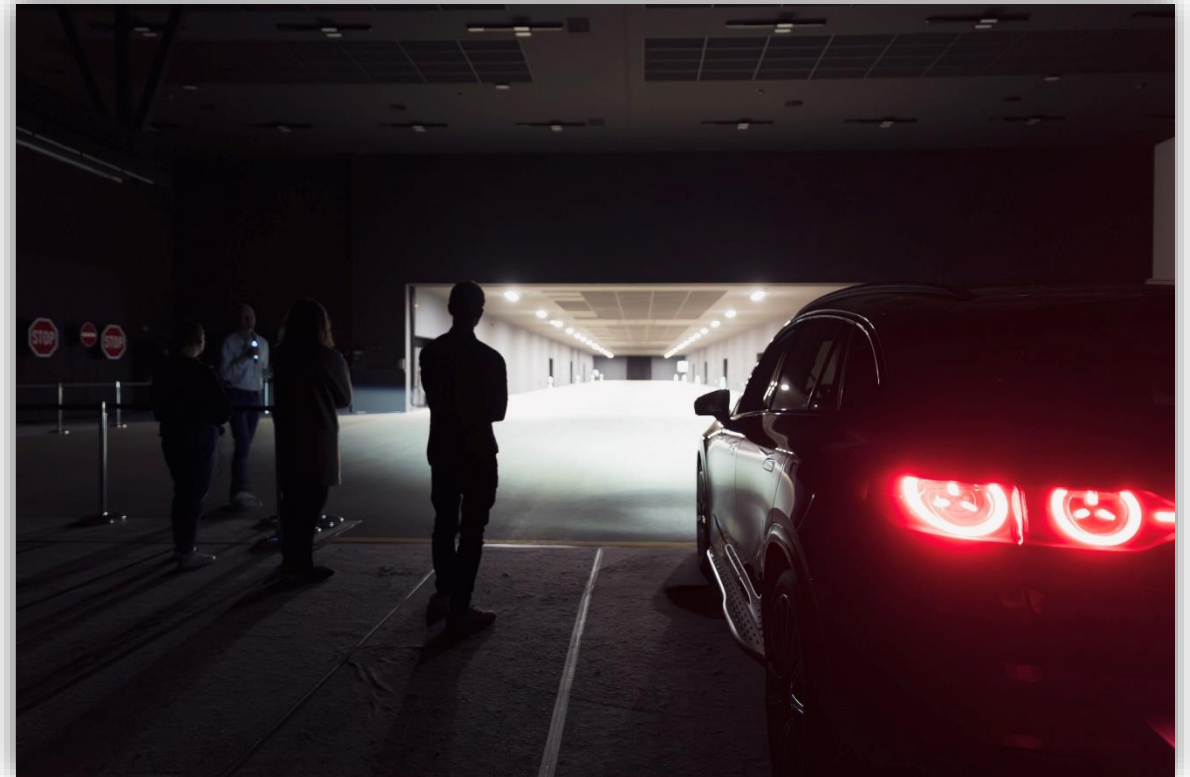
Location	Unterschleißheim, Germany
Physical Dimensions	132 meters in length; approx. 3,000 square meters in area
Primary Objective	Evaluating adaptive headlights, "light carpets," and entry lighting where computer simulations reach their limits

Surface: Aged Asphalt	A ground-grinded surface replicating a real road to evaluate the color and homogeneity of low-beam distributions
Surface: Footpath & Car Park	Specialized substrates used to analyze the clarity and sharpness of BMW light carpet signatures
Surface: Oiled Parquet	Replicates the unique reflective conditions of international trade fair vehicle presentations



Mercedes-Benz Light Testing Center

Feature	Technical Specification	Functional Purpose
Facility Length	135 meters (443 feet)	Allows for long-range beam distribution analysis and high-speed interaction testing.
Testing Capacity	Up to 5 vehicles simultaneously	Enables the simulation of complex traffic scenarios, including multiple oncoming and preceding vehicles.
Road Surface	Specialized Aged Asphalt mixture	Replicates the real-world reflective properties of older roads to evaluate beam homogeneity and color.
Environmental Control	Constant & Reproducible	Removes variables like weather and moon phases to ensure mathematical comparability between tests.
Key Simulation Tools	"Artificial Sun" mobile lights	Replicates low-sun conditions to test for HMI washout and sensor blinding/glare.
ADAS Integration	Rain and Spray simulation	Evaluates how moisture affects Adaptive Driving Beam (ADB) sensor recognition and light propagation.





Interior Lighting and the Human-Machine Interface (HMI)

Validation Domain	Methodology	Engineering Goal
Visual Ergonomics	Texture Enhancement	Use light angles to optimize perceived quality of materials (Leather/CFRP).
Optical Quality	Fluid Homogeneity	Ensure zero hotspots or color shifts across long LED contours.
HMI Legibility	Silverstein Function	Algorithmically calculate luminance based on background glare (Math-driven).
Safety Signaling	Active Color Coding	Green/Red ambient transitions signal Autonomous Control handover.
Driver Focus	Matrix Reading Lights	Directed beams isolate passengers to prevent driver distraction (Stray light control).



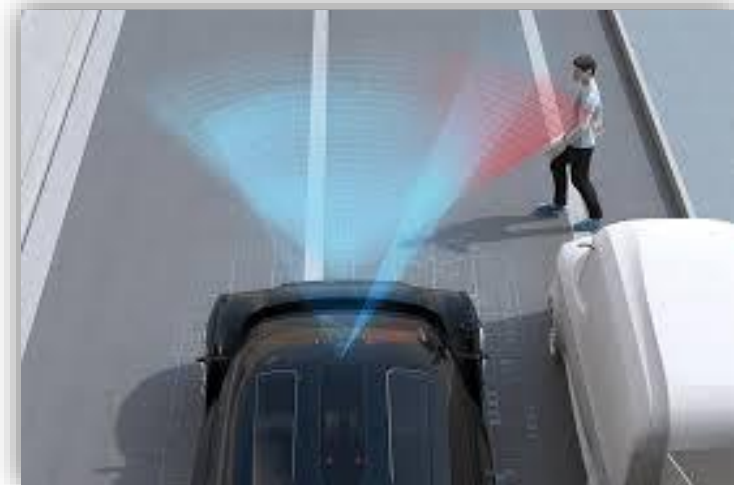
Proposed Scenarios for Assessment





EXTERIOR - HEADLAMPS

ID	Zone	Test Scenario	Method / Equipment	Procedure / Condition	Acceptance Criteria (Target)	Objective/Subjective Evaluation
1.1	Low Beam	Beam Uniformity & Cut-off	Digital Histogram (digital camera + software)	Park 10m from wall. ISO 100 photo. Analyze brightness graph in editor.	Histogram shows sharp drop at cut-off. No "jagged" peaks (hotspots) in dark zones.	O
1.2	Low Beam	Static Pedestrian Detection	Dummy (Cardboard + Dark Shirt)	Place low-reflectivity dummy on shoulder. Approach at 50km/h.	Driver identifies object at safe braking distance (>30-40m).	S





EXTERIOR - HEADLAMPS

ID	Zone	Test Scenario	Method / Equipment	Procedure / Condition	Acceptance Criteria (Target)	Objective/Subjective Evaluation
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1.3	Low Beam	Glare Check (Oncoming)	Visual / Partner	Observe from oncoming vehicle (driver eye height ~1.1m).	No dazzling glare to oncoming traffic. Cut-off stays below eye level.	Subjective
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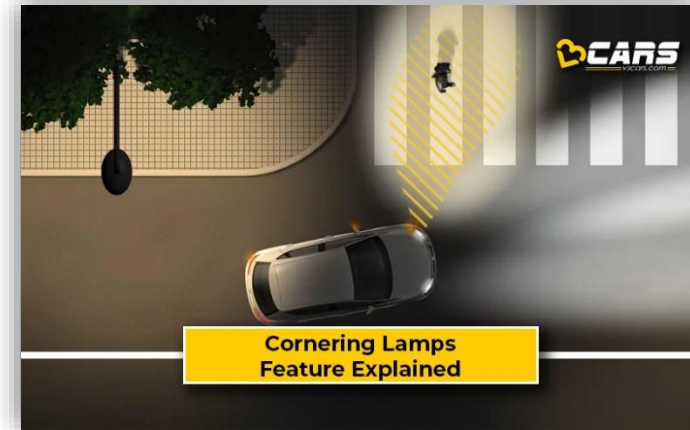
1.4	High Beam	Long Range Spot	Distance Markers	Place markers at 100m. Activate High Beam.	Lux reading (or visual clarity) at 100m is sufficient to identify road hazard.	Luxmeter required for this testing.
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EXTERIOR - ADAPTIVE & WEATHER

ID	Zone	Test Scenario	Method / Equipment	Procedure / Condition	Acceptance Criteria (Target)	Objective/Subjective Evaluation
2.1	Adaptive	Cornering Response	Long Exposure Photo (Tripod)	Drive through S-curve. Capture light trail on camera.	Light trail "leads" the vehicle. No lag between steering and light movement.	Objective:: 1) Width of cornering lamp 2) Angle when activates 3) Brightness



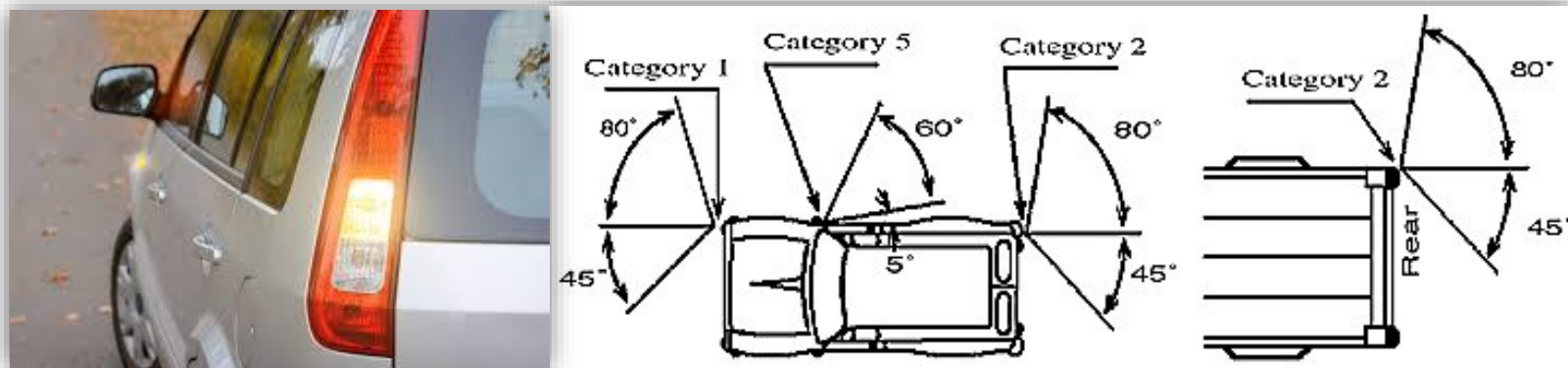
2.3	Reliability	Soiling Degradation	Mud/ Water Spray	Measure lux. Spray lens with dirt. Measure again.	Luminosity drops by <X% (Subjective: still usable). No erratic beam scattering.	Objective under conditions: = Create pattern stencils/stickers
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EXTERIOR - SIGNALING

ID	Zone	Test Scenario	Method / Equipment	Procedure / Condition	Acceptance Criteria (Target)
3.1	Signals	Off-Axis Visibility (Turn)	Visual / Protractor	Activate turn signal. Stand at 45° angle to the rear/front corner.	Blinking clearly visible. Light source not hidden by bodywork.





INTERIOR - DRIVER COMFORT (HMI)

ID	Zone	Test Scenario	Method / Equipment	Procedure / Condition	Acceptance Criteria (Target)
4.2	Screen	Min. Brightness (Fatigue)	Subjective / Biometric	Drive 20 mins in pitch black. Check eye strain/dryness.	Screen does not illuminate driver's face. No "after-image" when looking at road.





INTERIOR - UX & SCENARIOS

ID	Zone	Test Scenario	Method / Equipment	Procedure / Condition	Acceptance Criteria (Target)
5.4	Front Seat	The "Drop Test"	Black Object (Key/Phone)	Drop object into gap between seat/console.	Object visible via ambient or map light. No harsh shadows.



5.5	Trunk	Trunk Load Occlusion	2 Large Suitcases	Fill trunk. Check if light source is blocked.	Light source located <i>above</i> or <i>around</i> cargo line. Floor remains visible.
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UTILITY & SAFETY

ID	Zone	Test Scenario	Method / Equipment	Procedure / Condition	Acceptance Criteria (Target)
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6.1	Utility	Emergency Tire Change	Visual Check	Pitch black road. Hazard lights ON. Stand by wheel arch.	Enough peripheral light (hazards/puddle) to see wheel nuts.
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6.2	Utility	Welcome/ Approach	Visual Check	Unlock car from 10m away. Walk towards door.	Door handle and ground (puddle) illuminated before contact.
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Facility & Instrument Requirements





Dark Room to replicate these professional conditions.

Scope of Validation

- **Exterior:** Low-beam cut-off sharpness (Gradient analysis), static glare assessment, and mounting angle verification.
- **Interior:** HMI "Black Level" luminance testing (ISO 15008) in absolute darkness to validate driver comfort.



Hioki FT3424 (Light Meter/Datalogger)

ST T	Tên thiết bị	Đơn vị tính	SL	Đơn giá	Thành tiền	Tình trạng
1	Máy đo cường độ ánh sáng HIOKI FT3424  Model: FT3424 Hãng sản xuất: HIOKI Xuất xứ: Japan (Nhật Bản)	Cái	1.0	11,500,000	11,500,000 VND	Sẵn có

Thông tin chi tiết:

Giải đo:

20 lx: 0.00 lx đến 20.00 lx /1 count step

200 lx: 0.0 lx đến 200.0 lx

2000 lx: 0 lx đến 2000 lx

20000 lx: 0(0) lx to 2000(0) lx /10 count step

200000 lx: 0(00) lx~2000(00) lx /100 count step

Chế độ chọn giải: Tự động/ bằng tay

Độ tuyến tính: $\pm 2\%$ rdg. (Multiply by 1.5 for display values in excess of 3000 lx.)

Thời gian hồi đáp: Auto range: within 5 seconds, Manual range: within 2 seconds

D/A output: Output level: 2 V/range f.s. (2.5 V is output when the range f.s. is exceeded.)

Độ chính xác đầu ra: $\pm 1\%$ rdg. ± 5 mV (at display count)

Chức năng: Timer hold function, Memory function (Up to 99 measured data can be saved.), Hold, Auto power off function, Buzzer sound, Backlight, Zero adjustment

Nguồn: pin LR6 Alkaline battery x2 (continuous use 300 hours), Max. rated

Kích thước: 78 mm W x 170 mm H x 39 mm D, 310 g

Phụ kiện kèm theo: HDSDx1, pin LR6 Alkalinex2, túi đựng (soft) x1, dây đeo x1, Cáp USB (0.9 m/2.95 ft) x1, CD x1

EMIN
Thinking & Measuring Everything





Digital Angle Gauge / Level Box

2	Nivo cân máy điện tử tích hợp đo góc INSIZE 2170-1 (0-360 / 0.1°)  Model: 2170-1 Hãng sản xuất: INSIZE Xuất xứ: China	Cái	1.0	480,000	480,000 VND	8 - 10 tuần
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Thông tin chi tiết:

Made with Xodo PDF Reader and Editor

Cách sử dụng như thước thủy.
Khung máy bằng nhôm
Dải đo : 0 - 360 (90°x4)
Độ phân giải : 0.1°(=1.745mm/m)
Cấp chính xác : $\pm 0.1^\circ$
Chiều dài : 60mm
Nút ấn : ON/OFF, ZERO,
Pin : AAA
Tự động tắt nguồn





Laser Cross Line Level

1.2: "Place objects at **45°/60° angles**".

3.1: "Stand at **45° angle** to the rear/front corner".



CHỐT DEAL
NĂM MỚI

Xem 2
hình



Xem 5
nhận xét

Thông số
kỹ thuật



1 / 2

Máy cân mực laser Bosch GLL 3-80 (Tia đỏ)

★★★★★ (1 đánh giá)

Thương hiệu: Bosch | Máy cân mực Laser, Máy thủy bình Bosch

12.800.000đ -15.500.000đ -17% (Đã gồm VAT)

Trạng thái: ✓ Còn hàng, lấy hàng tại: 11 ngõ 4 Duy Tân, Cầu Giấy, Hà Nội

Chọn số lượng: Cho vào giỏ

ƯU ĐÃI



Tặng móc khóa Bosch (cho miền Bắc)
Đến khi hết ưu đãi



Đặt mua

Giao hàng tận nơi trên toàn quốc



Tư vấn

Chúng tôi sẽ gọi lại cho bạn

Trả góp

Duyệt nhanh, Không cần thế

Trả góp qua thẻ

Visa, Master, JCB, AmEX

Thường được mua cùng:



Chân máy thủy bình Bosch BT 160

910.000đ



Miễn phí giao hàng trong nội thành Hà Nội và nội thành TP. Hồ Chí Minh. [\(Xem thêm\)](#)



Được hàng trăm ngàn Doanh nghiệp tại Việt Nam lựa chọn: đầy đủ hóa đơn, hợp đồng, không chi phí ẩn [\(Xem thêm\)](#)



Bảo hành toàn quốc. [\(Xem trung tâm bảo hành\)](#)





Laser Rangefinder

3	Máy đo khoảng cách laze BOSCH GLM 100-25 C (100m)  Model: GLM 100-25 C Hãng sản xuất: BOSCH	Cái	1.0	5,249,537	5,249,537 VND	Sẵn có
Thông tin chi tiết: Khoảng cách đo: 0,08 - 100m Sai số: $\pm 1,5\text{mm}$ Camera: 5MP Laser: Đỏ Phạm vi đo độ nghiêng: 0 - 360 ° Thời gian đo lường: < 0,5 giây IP: IP54 Truyền dữ liệu: Bluetooth™ 4.2 Low Energy Dung lượng bộ nhớ (giá trị): 50 Pin: 3 x 1.5V LR6 AA Trọng lượng: 230g Kích thước: 176mm x 111mm x 77mm						





TES-137 (Luminance Meter)

2	Máy đo độ chiếu sáng TES TES-137	Cái	1.0	12,480,000	12,480,000 VND	Sẵn có
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Made with Xodo PDF Reader and Editor



Model: TES-137

Hãng sản xuất: TES

Xuất xứ: Taiwan (Đài Loan)



Thông tin chi tiết:

Hãng sản xuất: TES

Model: TES-137

Xuất xứ: Đài Loan

- Màn hình kép, màn hình LCD 4 chữ số
- Độ nhạy quang phổ gần đường cong CIE photopic.
- Đo mức độ từ 0,001 đến 1999k cd / m², từ 0,001 đến 580,0k fL.
- Đo tỷ lệ độ sáng A / B, độ lệch% A, khác nhau B-A và độ sáng đỉnh
- Các chức năng hiệu chỉnh màu sắc và tham số hiệu chuẩn sử dụng
- Đo độ chói tích lũy
- Chức năng Max / Min / AVG
- Chức năng so sánh
- Chức năng data HOLD
- Chức năng đọc và ghi nhớ dữ liệu
- Chức năng Data Logger
- Giao diện USB

Các ứng dụng:

- Đo độ sáng của màn hình LCD, CRT, đèn LED, vv
- Đo ánh sáng đường phố, ánh sáng đường hầm, vv
- Đo lường các cơ sở chiếu sáng sân bay.
- Đo truyền tải cho tấm phản cực LCD và các bộ lọc khác nhau.
- Đo ánh sáng y tế.
- Chiếu sáng bất thường của biển số xe ô tô.
- Độ độ chiếu sáng của các cơ sở chiếu sáng khác nhau, vv
- Màn hình: màn hình LCD, màn hình LCD 4 chữ số
- Phạm vi:
 - +9.999, 99.99, 999.9, 9999, 99.99k, 999.9k, 1999k cd / m²
 - +9.999, 99.99, 999.9, 9999, 99.99k, 580.0k fL
- + Tự động thay đổi (6 bước) (1 fL = 3,426 cd / m²)
- Góc đo: 2°
- Hiển thị quá phạm vi: hiển thị " OL "
- Độ phân giải: 0.001 cd / m², 0.001 fL
- Độ chính xác: +/- 3% rdg +/- 5dgt (Được hiệu chuẩn theo chuẩn, đèn sợi đốt, 2856 K, ở 25/77°)
- Dung lượng bộ nhớ dữ liệu: 200 bộ
- Dung lượng Data Logger : micro SD Card 4GB
- Tốc độ đo: xấp xỉ 5 lần / giây



Calibrite ColorChecker Passport Photo 2





Atmospheric Scattering Simulation (Beam Geometry)

- Optical **Haze Generator** for Beam Visualization
 - Needed for a **Beam Geometry Test** (checking where the light goes), not a **Penetration Test** (checking how far the driver can see in real weather).
 - Visualizes light path and 'leakage' (stray light) that is invisible in clear air. Critical for verifying *cut-off sharpness*

Antari Z-350 Fazer

+ **Antari FLG-5** (Heavy Fog Fluid)





Camera + Tripod

1.1: "Digital Histogram..."

2.1: "Long Exposure Photo..."



Voltage Stabilization Check – Digital Multimeter



Thank You





Appendix

Unit Definitions

- ❖ Lumen = Source Power
- ❖ Candela = Beam Intensity (Directional)
- ❖ Lux = Surface Illumination (What we measure)
- ❖ Nits = Eye Perception (Screen Glare)



Environmental Test Metric	Parameter Range	Industry Standard / Reference
Temperature Cycling	-40 °C to +100 °C	IEC 60068-2-14 (Thermal Shock)
Thermal Shock Transition	<6 seconds	Internal Highly Accelerated Life Testing (HALT)
High-Pressure Water Jet	Up to 100 bar at 80 °C	DIN 40050-9 / ISO 20653 (IP69K)
Vibration Frequency	10 Hz to 1,000 Hz	SAE J575 / ISO 16750-3 (Mechanical Loads)
Mechanical Shock	300 to 500 m/s ²	Shipping and impact simulation
Dust Medium	Unfired Portland Cement	SAE J575e Equivalent





Valeo – Headlight Testing

“Valeo's PictureBeam™ Monolithic technology manages each pixel independently to adjust intensity and flux with surgical precision”

Valeo/LucidDrive Simulation Tier	Method	Key Outputs
Conceptual Simulation	Shadow masks based on vehicle bounding boxes	Rapid angular resolution studies; idealized masking [7]
Physics-Based Simulation	Individual source pixel modeling in real-time	Prediction of stray light, ghosting, and Fresnel reflections [7]
Optical Optimization	CODE V error function constraints	Optimized lens thickness and image quality [7]
Regulatory Validation	FMVSS 108 track-test simulation	Quantitative glare assessment via sensor recordings [7]

[Smart car lights: PictureBeam™ Monolithic | Adaptive headlights | Valeo](#)



Marelli's "Pixel Rear Lamp" utilizes OLED-TFT technology

OLED Reliability Challenge	Mitigation Strategy	Validation Technique
Moisture/Oxygen Degradation	Thin-film encapsulation (TFE) / Hermetic sealing	Humidity endurance testing [10]
Thermal Aging	Advanced heat sinks / ACF bonding	K-factor junction temperature analysis [12, 13]
Non-Uniform Luminance	Pixel circuit compensation architectures	Homogeneity and aging uniformity tests [10]
Image Retention ("Burn-in")	Advanced driving schemes and feedback systems	Physics-based degradation modeling [10, 15]

